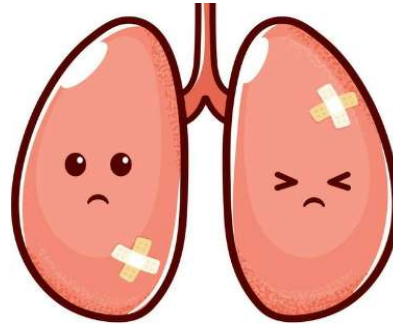


# PULMONARY EMBOLISM

A BLOOD CLOT THAT TRAVELS TO THE BLOOD VESSELS IN THE LUNGS BLOCKING BLOOD FLOW TO LUNG TISSUE — MAKING THE LUNGS SAD

## HEMODYNAMIC UNSTABLE PE S&S:

- Tachypnea w/  $\downarrow$  O<sub>2</sub> & vitals  $\Delta$ s (#1 S&S)
- Chest pain (often sudden onset)
- Hypotension
- Palpitations
- Diaphoresis
- Cough with or w/o hemoptysis
- Anxiety / Feeling of Impending Doom
- Confusion
- Tachycardia
- Syncope
- Dyspnea & SOB
- Sudden CP Arrest



## STABLE PE S&S:

- Cough, SOB, DOE, Tachypnea
- Pleuritic chest pain
- Tachycardia
- Anxiety
- Fever
- (pt may have any "unstable PE" S&S, but in milder presentation)

ALWAYS REMEMBER A STABLE PE HAS PROPENSITY TO BECOME UNSTABLE. CLOSE MONITORING & FOCUSED ASSESSMENT ARE

## UNSTABLE PE • NURSING INTERVENTIONS:

- Call "Rapid Response" Code
- Close monitoring, increased nursing vigilance
- ECG & Telemetry for dysrhythmias
- Hemodynamic monitoring (O<sub>2</sub>, BP, ABGs)
- BP Support — fluids, pressors, Strict Is&Os
- Pain & Anxiety Management
- O<sub>2</sub>, Respiratory Support



## STABLE PE NURSING INTERVENTIONS:

- Close monitoring
- ECG & Telemetry for dysrhythmias
- Bleeding assessment & risk mgmnt
- Vitals Support — BP, O<sub>2</sub>
- Pain & Anxiety Management
- Patient & Family Education

## COLLABORATIVE CARE & INTERVENTIONS:

- Stabilize Cardio-Pulmonary system
- Emergent Cardio & Pulmonary support
- Thrombolytic therapy — lyse the clot
- Surgical embolectomy via catheter or open Sx on cardiac bypass
- for Recurrent PE — IVC Filter & Thrombolytics

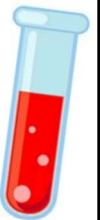


## COLLABORATIVE CARE & INTERVENTIONS:

- Heparin (unfractionated or LMWH) infusion or SQ injections
- Oral Anticoagulation
- Monitoring for therapeutic anticoagulation

## DIAGNOSTIC TESTING:

- LABS —  $\uparrow$  D-dimer, ( $>1.96$ mg/L), ABG analysis, PT, PTT, INR, CMP, CBC
- CXR — not often detected, may see infiltrates, atelectasis, or  $\uparrow$  of diaphragm or pleural effusion on affected side
- CT Scan — may detect PE (often w/IV contrast)
- V/Q (Ventilation/Perfusion) Scan — shows circulation of air & blood in lungs, can detect PE, uses IV contrast



## RISK FACTORS & VIRCHOW'S TRIAD:

Endothelial Injury + Hypercoagulable State + Venous Stasis = BLOOD CLOT

## OK, But what causes these things?

**Endothelial Damage:** Trauma to injury (incl venipuncture); indwelling devices (stents, valves, IV access); Medications; IV drug usage  
**Venous Stasis** — circulation is impaired due to: Immobilization (travelling long hours); Varicose veins; Surgery (esp hip or knee); Obstruction (late pregnancy, Obesity); Heart Failure; A-fib  
**Hypercoagulable State** — Dehydration; Oral birth control use; Post-partum; Severe illness (Sepsis); HIT (Heparin Induced Thrombocytopenia)

platelets & fibrin



## PATHOPHYSIOLOGY OF A BLOOD CLOT:

Intravascular Endothelial Injury > Activated Platelets gather > Clotting factors are released > Fibrin is created & comes in to hold everything together > More blood cells show up to "help" but keep collecting & a clot is formed & keeps growing larger.